

SIPLE DOME, Antarctica

WMO: 893450

Lat: 81.653S

Lon: 148.990W

Elev: 620

StdP: 94.10

Time Zone: -10.00 (W10)

Period: 97-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-49.3	-46.9	N/A	N/A	N/A	N/A	N/A	N/A	63.9	-33.1	63.2	-33.0	4.9	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.7	-0.9	N/A	-1.9	N/A	-4.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 62.3	38.6	34.0	DB	-52.0	5.0	3.7	3.5	-54.7	7.5	-56.8	9.5	-58.8	11.5	-61.5	14.0	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-22.8	-10.2	-16.1	-22.8	-26.7	-28.5	-27.7	-30.8	-32.5	-29.8	-23.4	-14.7	-9.8	(6)	
	DBStd	10.06	3.92	4.91	5.91	6.54	7.50	7.97	7.83	8.18	7.75	5.63	4.41	3.54	(7)	
	HDD10.0	11968	625	730	1018	1102	1193	1132	1264	1318	1195	1037	742	613	(8)	
	HDD18.3	15009	883	963	1276	1352	1451	1382	1522	1577	1445	1295	992	871	(9)	
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	8.1	6.9	7.1	7.5	10.5	7.9	8.6	9.3	9.1	9.6	7.3	6.6	6.2	(14)	
(15) Precipitation	PrecAvg	245	15	12	17	20	25	29	29	27	26	20	13	12	(15)	
	PrecMax	373	34	29	69	74	61	60	96	66	70	34	51	44	(16)	
	PrecMin	93	2	1	2	2	3	3	2	5	2	2	2	2	(17)	
	PrecStd	71	9	8	14	16	15	16	22	15	18	9	11	10	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	-2.1	-5.8	-9.2	-8.6	-9.3	-12.0	-13.7	-9.2	-6.1	-1.7	2.5	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	-0.8	-5.4	-9.2	-12.6	-11.3	-12.5	-14.5	-16.9	-12.4	-10.4	-4.9	-0.7	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	-1.7	-7.1	-12.2	-14.8	-15.4	-13.8	-17.1	-18.2	-15.4	-13.3	-6.2	-1.7	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	-4.8	-8.9	-14.3	-17.2	-17.6	-16.6	-19.9	-20.8	-18.2	-15.1	-8.4	-4.8	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.2	8.5	8.6	8.7	8.7	8.6	8.5	9.4	9.7	8.2	7.7	(35)	
		MCDBR	10.7	10.8	8.9	9.9	10.3	9.5	10.3	9.0	10.2	11.6	10.9	11.2	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.215	0.193	0.146	N/A	N/A	N/A	N/A	N/A	0.138	0.180	0.199	0.212	(40)	
	taud		2.219	2.218	2.066	N/A	N/A	N/A	N/A	N/A	2.019	2.203	2.243	2.234	(41)	
	Ebn at Noon		962	887	687	0	0	0	0	0	681	907	986	996	(42)	
	Edh at Noon		92	67	34	0	0	0	0	0	34	68	89	97	(43)	
(44) All-Sky Solar Radiation	RadAvg		8.21	4.70	1.15	0.00	0.00	0.00	0.00	0.00	0.50	3.33	7.64	9.58	(44)	
	RadStd		0.22	0.15	0.05	0.00	0.00	0.00	0.00	0.00	0.02	0.10	0.26	0.25	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page